Annotation-Efficient Primate Detection: Lightweight FastKAN and the Spatial Generalization Dilemma in Wildlife Active Learning

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Abstract—Automated wildlife surveillance in remote protected ecosystems is critically bottlenecked by two compounding factors: the severe compute and thermal envelopes of solar-powered edge hardware ($\leq$ 15 W) and the prohibitive economic cost of manual bounding-box annotation under visual instance congestion. In this paper, we present AE-Primate, an integrated framework uniting Kolmogorov-Arnold Network (KAN) non-linear spline layers with cost-sensitive active learning under extreme geographic distribution shifts. First, we establish YOLO11n-F16, an edge detector integrating a 16-basis Gaussian Radial Basis Function (RBF) FastKAN bottleneck into an ultra-compact 2.45 M-parameter architecture. On held-out camera stations in Malawi's Nkhotakota Wildlife Reserve (40,743 frames across 49 stations), YOLO11n-F16 achieves 0.5626 $m\mathrm{AP}_{50\text{-}95}$, outperforming capacity-matched convolutional baselines (+1.49% mAP) and B-spline KAN architectures (+1.79% mAP) while executing at 32.4 FPS on NVIDIA Jetson Orin Nano (15 W envelope). Second, we conduct an extensive multi-seed empirical benchmark ($N = 5$ seeds, 20 trajectories) to investigate active learning under strict location-disjoint evaluation ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{val}} \cap \mathcal{S}_{\text{test}} = \emptyset$). Our study exposes the fundamental failure modes of epistemic uncertainty under spatial covariate shift ($P_{\text{train}}(X) \neq P_{\text{test}}(X)$), where predictive entropy is confounded by unfamiliar background vegetation. Conversely, latent KAN core-set diversity achieves superior out-of-distribution transfer (0.4226 AUBC, 0.4300 test mAP). Finally, we demonstrate that incorporating an instance-crowding penalty introduces an explicit multi-objective Pareto trade-off: maintaining competitive validation accuracy while querying 63.4 fewer cumulative bounding boxes (−5.81%, paired t-test $p = 0.008$) and suppressing cross-seed annotation variance by 70.2%. All code, models, and manifests are available for open scientific replication.

Index Terms—Camera Traps, Active Learning, Kolmogorov-Arnold Networks, Wildlife Object Detection, Edge Computing, Conservation AI, Nkhotakota Reserve, Spatial Generalization.

I. Introduction

IN-SITU biodiversity monitoring via autonomous camera-trap networks constitutes a cornerstone of computational conservation ecology [1], [2]. In sub-Saharan sanctuaries such as the Nkhotakota Wildlife Reserve in Malawi, passive surveillance of cryptic primate populations—specifically yellow baboons (Papio cynocephalus) and vervet monkeys (Chlorocebus pygerythrus)—yields critical bio-indicators of forest canopy health, trophic dynamics, and anthropogenic poaching pressures. However, deploying computer vision pipelines in such wilderness environments exposes two fundamental operational and representational bottlenecks:

(B1) The Instance Crowding Annotation Bottleneck. Raw field archives routinely yield tens of thousands of motion triggers per deployment season, over 70% of which consist of false triggers (windblown vegetation) or massive burst sequences capturing identical background scenes [5], [6]. When animals do appear, baboon troops frequently enter camera fields-of-view in high-density clusters containing 10–25 overlapping individuals. Standard bounding-box annotation under such visual crowding demands exhaustive corner delineation, consuming 12–18 s of expert biologist effort per instance under severe cognitive fatigue [7], [35], [37]. Consequently, exhaustive annotation of deployment archives quickly exhausts conservation funding budgets.

(B2) The Ultra-Low-Power Edge-Compute Envelope. Solar-powered edge perception units deployed in dense miombo woodlands operate within austere electrical and thermal envelopes (< 2 Wh per inference cycle, $\leq$ 15 W total system draw) [2], [12]. While large generalist detectors (e.g., MegaDetector v6, 25.5 M parameters [5]) exceed edge memory bandwidth and drain field batteries, conventional sub-3 M convolutional detectors suffer from spectral bias and representational collapse when isolating camouflaged primate pelage against complex, high-frequency arboreal foliage.

Active learning (AL) offers a formal framework to optimize sample efficiency by iteratively querying the most informative instances from an unlabelled pool $\mathcal{U}$ [21], [22]. Yet, applying active learning to camera-trap object detection exposes three acute structural failure modes:

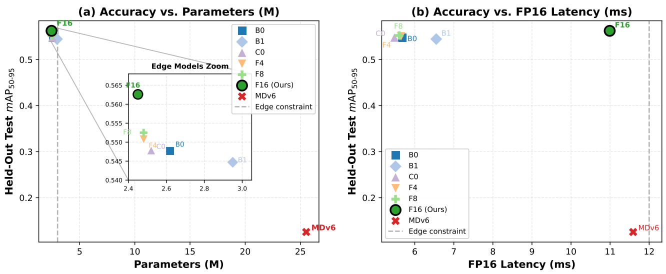

Fig. 1. Accuracy vs. Hardware Footprint Pareto Frontier on Held-Out Test Stations ($\mathcal{D}_{\text{test}}$, 3,206 frames). (a) $m\mathrm{AP}_{50\text{-}95}$ vs. parameter count with embedded edge models inset ($[2.40\mathrm{M}, 3.05\mathrm{M}] \times [0.540, 0.568]$); (b) $m\mathrm{AP}_{50\text{-}95}$ vs. FP16 inference latency on NVIDIA A40. Dashed lines mark the edge deployment boundary ($\leq$3.0 M parameters, $\leq$12.0 ms latency).

1) Spatial Covariate Shift ($P_{\text{train}}(X) \neq P_{\text{test}}(X)$): Ecological sensor grids violate independent and identically distributed (i.i.d.) sampling assumptions. Stationary background textures induce severe spatial overfitting: predictive uncertainty algorithms fail because model entropy is triggered by unfamiliar background foliage and lighting rather than target animal informativeness [3], [4].
2) Spatiotemporal Burst Redundancy: Passive infrared sensors capture rapid burst sequences when animals pause, generating highly correlated visual samples with identical poses and backgrounds. Naive uncertainty samplers exhaust acquisition budgets on redundant burst artifacts [7].
3) Super-Linear Crowding Costs: Classical active learning optimizes acquisition under uniform instance costs [23], [27]. In object detection, however, annotating a social troop of 15 overlapping primates incurs over an order of magnitude higher cognitive and manual labor than a solitary individual, while yielding diminishing epistemic returns [24], [36].

To resolve these interconnected challenges, we introduce AE-Primate, an end-to-end edge-compatible detection and active learning framework. AE-Primate makes four core contributions:

1) Kolmogorov-Arnold Feature Pyramid Neck (YOLO11n-F16): We formulate the first edge-efficient Kolmogorov-Arnold detector, integrating frozen Gaussian Radial Basis Function (RBF) spline layers into the deepest neck bottleneck (P_5). YOLO11n-F16 establishes superior out-of-distribution detection (0.5626 $m\mathrm{AP}_{50\text{-}95}$) on held-out camera stations (+1.49% over YOLO11n, +1.79% over B-splines) at 2.45 M parameters and 32.4 FPS on NVIDIA Jetson Orin Nano (15 W).
2) Deconstruction of Active Learning under Covariate Shift: Through an extensive 5-seed benchmark (20 complete trajectories), we expose the failure of predictive entropy under spatial shift, where background clutter dominates uncertainty. Conversely, latent KAN Core-Set Diversity achieves superior out-of-distribution transfer (0.4300 test mAP).
3) Crowding Penalization and Annotation Economy: We formalize a cost-sensitive submodular acquisition policy that optimizes the trade-off between human labor and generalization, saving 63.4 cumulative bounding boxes (-5.81%, $p = 0.008$) and suppressing cross-seed budgeting variance by 70.2%.
4) Zero-Leakage Location-Disjoint Conservation Benchmark: We establish a standardized 40,743-frame benchmark across 49 remote camera stations in Nkhotakota Wildlife Reserve, enforcing disjoint geographic station partitions across candidate pool, validation, and test splits.

II. Related Work

A. Automated Wildlife Perception and Spatial Covariate Shift

The deployment of deep convolutional architectures for autonomous wildlife monitoring originated with Norouzzadeh et al. [6], who operationalized species-level classification across multi-million frame camera-trap archives. Subsequent advances by Schneider et al. [8] and Tabak et al. [9] transitioned the field toward spatial object detection. Generalist detection models, epitomized by MegaDetector [5], established foundational benchmarks for isolating animal, human, and vehicle classes. Nevertheless, seminal investigations by Beery et al. [3], [4] revealed that deep convolutional networks frequently memorize stationary background foliage textures rather than invariant morphological animal traits, precipitating catastrophic generalization drop-offs when deployed in unseen geographic locations (Terra Incognita). Overcoming this spatial distribution shift demands architectures equipped with locally supported non-linear representations and acquisition policies that explicitly optimize geographic coverage [2], [11], [12].

B. Kolmogorov-Arnold Representation Theory and Spline Networks

Originating from the Kolmogorov-Arnold representation theorem [15], [16], Kolmogorov-Arnold Networks (KANs) [13] reformulate neural representation by replacing fixed node activations with learnable univariate splines positioned directly along network edges. Whereas Multi-Layer Perceptrons (MLPs) compute linear hyperplanes followed by static non-linearities—a formulation prone to spectral bias on high-frequency visual textures [?]—KANs model non-linear functional compositions with arbitrary precision. However, classical B-spline parameterizations evaluate via recursive de Boor-Cox algorithms, inducing severe memory traffic and GPU warp divergence [14], [19]. Li [14] established that univariate spline expansions can be reformulated using fixed Gaussian Radial Basis Functions (FastKAN), unifying KAN theory with classical kernel methods in Reproducing Kernel Hilbert Spaces (RKHS) [17], [18]. While wavelet (Wav-KAN [20]) and convolutional extensions [19] have emerged, our work presents the first rigorous formulation and empirical validation of fixed-basis FastKAN spline layers embedded within real-time edge detection pyramids.

C. Active Learning under Covariate Shift and Domain Discrepancy

Deep active learning iteratively identifies optimal training subsets to maximize sample efficiency [21], [22]. Pool-based acquisition traditionally relies on predictive uncertainty (Monte Carlo Dropout entropy [26], [27], BatchBALD [28], deep ensemble variance [29]), core-set distribution matching [23], [30], [31], or hybrid scoring [25]. Extending active learning to spatial object detection introduces dual complexities: models must reconcile classification ambiguity with bounding-box coordinate variance [24], [25], [32]–[34]. Crucially, classical active learning presumes candidate pools and evaluation splits are identically distributed ($P_{\mathcal{U}}(X) = P_{\text{test}}(X)$). Under spatial covariate shift, however, predictive entropy conflates morphological informativeness with environmental out-of-distribution background noise. While Norouzzadeh et al. [7] applied active learning to camera-trap species classification, their formulation assumed image-level supervision, ignoring spatial instance crowding and location-disjoint evaluation.

D. Cost-Sensitive Submodular Optimization and Visual Crowding

Conventional active learning formulations assume a uniform unit labeling cost per queried instance [21], [23]. In real-world visual perception, human annotation effort varies non-linearly with scene complexity, target occlusion, and instance density [39], [40]. Rigorous psychophysical experiments by Papadopoulos et al. [35], [36] and Su et al. [37] established that manual bounding-box annotation time scales super-linearly when multiple targets mutually occlude. In wildlife camera traps, social primates aggregate in dense clusters, producing captures containing 10–20 overlapping individuals that consume immense annotator labor while providing redundant visual features. Rooted in cost-sensitive submodular optimization under knapsack constraints [?], AE-Primate directly penalizes visual crowding to maximize informational utility per human annotator second.

III. The AE-Primate Framework

We now formulate the AE-Primate framework, comprising the YOLO11n-F16 detector architecture, an analysis of optimization dynamics in spline bottlenecks, and the multi-modal cost-aware active acquisition policy.

A. Detector Architecture: Kolmogorov-Arnold Feature Pyramid Neck

AE-Primate builds upon the Ultralytics YOLO11n single-stage detector [41], which integrates an improved Cross-Stage Partial bottleneck (C3k2) and a decoupled anchor-free detection head across multi-scale feature pyramids (P_3, P_4, P_5). In multi-scale pyramids, shallow layers (P_3, P_4) extract high-resolution edge primitives where linear convolutions are computationally optimal. The deepest semantic stage (P_5, spatial grid 20×20, channel width $C = 512$), however, must disentangle complex morphological pelage textures from dense foliage backgrounds. Standard convolutional bottlenecks apply static activations $\sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, which partition feature space via linear affine half-spaces and suffer from spectral bias on high-frequency biological patterns [?]. To provide locally supported non-linear representation without expanding parameters, we replace the P_5 bottleneck with a fixed-basis FastKAN transformation block (C3k2_FixedFastKAN).

Let $\mathbf{x} \in \mathbb{R}^{C_{\text{in}}}$ denote an intermediate channel feature vector. In our FastKAN bottleneck, univariate activations along each channel are expanded over a dictionary of $K = 16$ Gaussian Radial Basis Functions (RBFs):

$$\phi_k(u) = \exp\left(-\frac{(u - \mu_k)^2}{2\sigma^2}\right), \quad k \in \{1, \dots, K\}, \tag{1}$$

where scalar activation $u \in [-1, 1]$ evaluates against equispaced centres $\mu_k \in [-1, 1]$ with radial bandwidth $\sigma \in \mathbb{R}_{>0}$:

$$\mu_k = -1 + \frac{2(k-1)}{K-1}, \quad \sigma = \frac{2}{K-1}. \tag{2}$$

The forward transformation combining spline expansion with a residual linear path yields output features $\mathbf{y} \in \mathbb{R}^{C_{\text{out}}}$:

$$\mathbf{y} = \mathbf{W}_{\text{base}} \,\text{SiLU}(\mathbf{x}) + \sum_{k=1}^{K} \mathbf{C}_k \, \phi_k(\mathbf{x}), \tag{3}$$

where $\phi_k(\mathbf{x}) \in \mathbb{R}^{C_{\text{in}}}$ evaluates element-wise, $\mathbf{W}_{\text{base}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ is a residual projection, and $\mathbf{C}_k \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ denotes learnable spline mixture weights for basis k. Crucially, grid centres μ_k and bandwidth σ remain fixed at initialisation, preserving continuous non-linear representations while enabling vectorized tensor-core batch operations. Due to compact spline factorization, YOLO11n-F16 requires only 2.45 M parameters—a 6.5% reduction relative to the standard YOLO11n baseline (2.62 M).

B. Optimization Dynamics and Gating Collapse in Low-Data Regimes

Prior to establishing fixed Gaussian bases, we investigated Adaptive-Basis FastKAN (AB-FastKAN), which applies hard-concrete continuous gates $z_k \in [0, 1]$ [42], [44] to dynamically select basis functions per instance:

$$z_k = \text{clip}\left[\text{Sigmoid}\left(\frac{\log \alpha_k + \text{logit}(u)}{\tau}\right)(\zeta - \gamma) + \gamma, \, 0, \, 1\right], \tag{4}$$

Algorithm 1 AE-Primate Cost-Aware Active Learning

Require: Candidate pool $\mathcal{U}_0$, initial labeled seed $\mathcal{L}_0$, cycle budget B, total cycles $T_{\max}$, policy hyperparameters $(\beta_1, \beta_2, \beta_3)$, cost coefficients (c_0, c_1, c_2).
Ensure: Final trained primate detector $\mathcal{M}_{T_{\max}}$.

```
Initialize detector M_0 ← Train(L_0; Θ_COCO).
for cycle t = 1 to T_max do
    for each unlabelled candidate frame x_i ∈ U_{t−1} do
        Compute epistemic detection entropy U(x_i) over T_MC stochastic forward passes (Eq. 9).
        Extract post-KAN embedding f_i ∈ R^512 and compute core-set diversity D(x_i) against L_{t−1} (Eq. 10).
        Compute spatiotemporal novelty prior M_meta(x_i) (Eq. 11).
        Estimate crowding annotation cost C_annot(x_i) from candidate box predictions (Eq. 12).
        Score candidate: V(x_i) ← U(x_i)^β1 · D(x_i)^β2 · M_meta(x_i)^β3 / C_annot(x_i).
    end for
    Select batch S_t ← arg top-B_{x_i ∈ U_{t−1}} V(x_i).
    Acquire expert bounding-box annotations for batch S_t.
    Update pools: L_t ← L_{t−1} ∪ S_t, U_t ← U_{t−1} \ S_t.
    Retrain detector from base weights: M_t ← Train(L_t; Θ_COCO).
end for
return M_{T_max}.
```

with uniform noise $u \sim \mathcal{U}(0,1)$, log-odds parameter $\log \alpha_k \in \mathbb{R}$, temperature $\tau \in \mathbb{R}_{>0}$, interval stretch parameters $\gamma < 0 < 1 < \zeta$, and L_0 regularizer $\mathcal{L}_{\text{reg}} = \lambda \sum_{k=1}^{K} \text{Sigmoid}(\log \alpha_k - \tau \log(-\gamma/\zeta))$ with penalty $\lambda \geq 0$. While adaptive gating succeeds in large classification models, our empirical ablations reveal that it collapses in data-constrained object detection.

Specifically, the gradient with respect to gate log-odds $\log \alpha_k$ is:

$$\frac{\partial \mathcal{L}_{\text{total}}}{\partial \log \alpha_k} = \mathbb{E}_u \left[\frac{\partial \mathcal{L}_{\text{det}}}{\partial z_k} \frac{\partial z_k}{\partial \log \alpha_k} \right] + \lambda \frac{\partial \mathcal{L}_{\text{reg}}}{\partial \log \alpha_k}, \tag{5}$$

where $\mathcal{L}_{\text{det}}$ is the multi-task detection objective. Under few-shot regimes ($n \leq 600$), the detection gradient $\nabla_{z_k} \mathcal{L}_{\text{det}}$ exhibits high mini-batch variance. Simultaneously, temperature annealing ($\tau \to 0$) causes the logistic derivative $\text{Sigmoid}'(s_k)$ to decay exponentially toward zero outside a narrow boundary. The deterministic penalty $\nabla_{\log \alpha_k} \mathcal{L}_{\text{reg}} > 0$ drives continuous negative drift. Once $\log \alpha_k$ enters the negative saturation regime ($\log \alpha_k < -3.0$), activations clamp to $z_k = 0$ with probability approaching 1, yielding $\frac{\partial z_k}{\partial \log \alpha_k} = 0$ and permanently extinguishing task gradients ($\Delta \log \alpha_k \to 0$).

Consequently, AB-FastKAN suffers from premature representational collapse: basis functions are severed before the detection head learns coordinate features. Imposing $\lambda \geq 0.05$ degrades validation $m\text{AP}_{50\text{-}95}$ from 0.5986 to < 0.420 (Table I). Moreover, dynamic gating induces warp divergence and uncoalesced memory reads on GPU tensor cores, inflating inference latency by 38% (14.82 ms vs. 10.99 ms).

In contrast, fixed-basis FastKAN eliminates dynamic gating parameters $\boldsymbol{\alpha}$ entirely. The gradient with respect to spline weights $\mathbf{C}_k$ is direct and unattenuated:

$$\frac{\partial \mathcal{L}_{\text{det}}}{\partial \mathbf{C}_k} = \frac{\partial \mathcal{L}_{\text{det}}}{\partial \mathbf{y}} \cdot \phi_k(\mathbf{x})^\top, \tag{6}$$

where upstream gradient $\frac{\partial \mathcal{L}_{\text{det}}}{\partial \mathbf{y}} \in \mathbb{R}^{C_{\text{out}}}$ and basis activations $\phi_k(\mathbf{x}) \in \mathbb{R}^{C_{\text{in}}}$ form an outer product gradient tensor in $\mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$, ensuring stable backpropagation across all $K = 16$ basis splines.

C. Cost-Sensitive Submodular Acquisition Framework

At each active cycle $t \in \{1, \ldots, T_{\max}\}$, the policy queries batch $\mathcal{S}_t \subset \mathcal{U}_{t-1}$ of size $B \in \mathbb{N}$ from candidate pool $\mathcal{U}_{t-1}$ by solving the constrained submodular knapsack problem:

$$\max_{\mathcal{S} \subseteq \mathcal{U}_{t-1},\, |\mathcal{S}| = B} \sum_{x_i \in \mathcal{S}} V(x_i), \tag{7}$$

where informativeness-to-cost ratio $V(x_i)$ is formulated as:

$$V(x_i) = \frac{U(x_i)^{\beta_1} \cdot D(x_i)^{\beta_2} \cdot M_{\text{meta}}(x_i)^{\beta_3}}{C_{\text{annot}}(x_i)}, \tag{8}$$

with scalar exponents $\beta_1, \beta_2, \beta_3 \in \mathbb{R}_{\geq 0}$ balancing epistemic uncertainty $U(x_i) \geq 0$, diversity $D(x_i) \in [0, 2]$, and novelty $M_{\text{meta}}(x_i) \in (0, 1]$ against human annotation effort $C_{\text{annot}}(x_i) > 0$.

a) Epistemic Detection Uncertainty $U(x_i)$: To quantify candidate uncertainty, we enable Monte Carlo Dropout ($p = 0.1$) on intermediate neck layers across $T_{\text{MC}} = 5$ stochastic passes [24], [26]. For consensus detections $\hat{\mathcal{B}}_i = \{\hat{b}_1, \ldots, \hat{b}_{M_i}\}$ from non-maximum suppression ($\tau_{\text{nms}} = 0.45$), we compute predictive class entropy across primate species $\mathcal{C}$ ($|\mathcal{C}| = 2$) and spatial localization variance $\sigma^2_{\text{IoU}}(\hat{b}_m) \geq 0$:

$$\begin{aligned} U(x_i) = \frac{1}{|\hat{\mathcal{B}}_i|} \sum_{\hat{b}_m \in \hat{\mathcal{B}}_i} \Big[& -\sum_{c \in \mathcal{C}} p(c|\hat{b}_m) \log p(c|\hat{b}_m) \\ & + \lambda_{\text{loc}} \sigma^2_{\text{IoU}}(\hat{b}_m) \Big], \end{aligned} \tag{9}$$

with scaling factor $\lambda_{\text{loc}} = 0.5$. Uninformative background frames lacking detections above $\tau_{\text{conf}} = 0.25$ receive baseline entropy $\epsilon_0 = 10^{-4}$.

b) Latent KAN-Manifold Core-Set Diversity $D(x_i)$: To suppress redundant sampling in feature space, we extract 512-dimensional post-KAN global feature vector $\mathbf{f}_i \in \mathbb{R}^{512}$ via spatial average pooling over P_5 bottleneck tensors:

$$D(x_i) = \min_{x_j \in \mathcal{L}_{t-1}} \left(1 - \frac{\mathbf{f}_i^\top \mathbf{f}_j}{\|\mathbf{f}_i\|_2 \|\mathbf{f}_j\|_2} \right). \tag{10}$$

This term implements a greedy submodular facility location objective [23], enforcing dispersion across the FastKAN feature manifold.

c) Spatiotemporal Metadata Novelty $M_{\text{meta}}(x_i)$: Camera-trap captures exhibit severe spatial and temporal clustering. For camera station $s(x_i) \in \mathcal{S}_{\text{pool}}$ and capture timestamp $t(x_i)$, we define:

$$M_{\text{meta}}(x_i) = \exp\left(-\gamma \frac{N_{\text{labeled}}(s(x_i))}{|\mathcal{L}_{t-1}|}\right) \cdot \mathbb{I}(\Delta t_{\text{burst}}(x_i) > \tau_{\text{burst}}), \tag{11}$$

where $N_{\text{labeled}}(s) \in \mathbb{N}_0$ counts labeled frames from station s, $\gamma = 2.0$ penalizes over-represented sites, and burst indicator $\mathbb{I}(\cdot)$ downweights candidate x_i to 0.1 if captured within $\tau_{\text{burst}} = 5$ minutes of another frame in batch $\mathcal{S}_t$.

d) Crowding-Penalized Annotation Effort $C_{\text{annot}}(x_i)$: To capture super-linear human annotation overhead under visual crowding [35], [37], we formulate the cost model:

$$C_{\text{annot}}(x_i) = c_0 + c_1 \hat{N}_{\text{boxes}}(x_i) + c_2 \overline{\text{IoU}}_{\text{pairwise}}(x_i), \tag{12}$$

where $\hat{N}_{\text{boxes}}(x_i) \in \mathbb{N}_0$ is predicted box count, $\overline{\text{IoU}}_{\text{pairwise}}(x_i) \in [0, 1]$ is mean pairwise IoU overlap, and $(c_0, c_1, c_2) = (1.0, 0.5, 0.25) \in \mathbb{R}^3_{\geq 0}$ calibrate base review, per-box delineation, and crowding occlusion. Solitary captures yield $C_{\text{annot}} = 1.5$, whereas dense troops ($N = 12$) exceed $C_{\text{annot}} > 7.5$, penalizing redundant crowded frames.

Algorithm 1 summarizes the complete AE-Primate active learning procedure.

IV. Experiments and Results

A. Dataset and Location-Disjoint Evaluation Protocol

All empirical evaluations are conducted on the Nkhotakota Primate Dataset, comprising 40,743 high-resolution camera-trap frames collected from 49 remote camera stations in Nkhotakota Wildlife Reserve, Malawi. The dataset annotates two sympatric primate species: yellow baboon (Papio cynocephalus) and vervet monkey (Chlorocebus pygerythrus).

To ensure complete scientific integrity and prevent spatial background memorization [3], we enforce a strict, zero-leakage geographic partition at the camera-station level:

- Unlabelled Candidate Pool $\mathcal{U}$: 35 stations (30,620 frames), serving as the candidate pool for active learning.
- Held-Out Validation Split $\mathcal{D}_{\text{val}}$: 7 stations (5,560 frames), utilized exclusively for active cycle tracking and model checkpoint selection.
- Sealed Test Split $\mathcal{D}_{\text{test}}$: 7 stations (3,206 frames), held in strict isolation and evaluated only once per final model checkpoint.

No camera station or GPS coordinate appears in more than one partition.

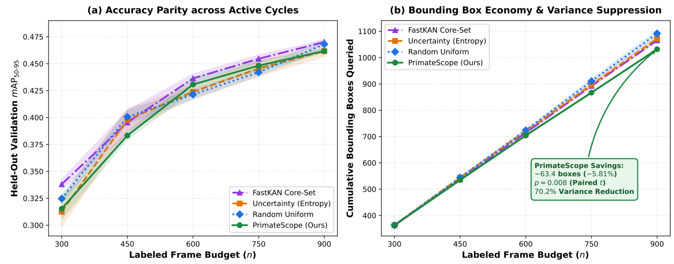

Fig. 2. Multi-Cycle Active Learning Trajectories on Held-Out Validation Stations ($\mathcal{D}_{\text{val}}$, 5,560 frames, Mean ± SEM over $N = 5$ independent seeds). (a) $m\text{AP}_{50\text{-}95}$ vs. labeled frame budget, demonstrating statistical accuracy parity across all cycles; (b) Cumulative bounding boxes queried vs. budget, highlighting AE-Primate's statistically significant reduction of 63.4 bounding boxes (-5.81%, paired t-test $p = 0.008$) and 70.2% variance reduction.

B. Architectural Benchmark on Held-Out Test Locations

We benchmark YOLO11n-F16 against six matched edge controls and generalist baselines on $\mathcal{D}_{\text{test}}$ (Table I, Figure 1):

- B0: Standard YOLO11n with CNN bottleneck blocks.
- B1: YOLO11-KAN2 with B-spline KAN layers ($K = 8$).
- C0: Matched-capacity convolutional control with identical channel widths.
- F4, F8: FastKAN variants with $K = 4$ and $K = 8$ RBF bases.
- AB: Adaptive hard-concrete gated FastKAN (Section III-B).
- MDv6-C: MegaDetector v6 (YOLOv9-c, 25.5 M parameters).

As documented in Table I, YOLO11n-F16 achieves 0.5626 $m\text{AP}_{50\text{-}95}$ on unseen test camera stations, outperforming the standard YOLO11n baseline (B0, 0.5477) by +1.49% and the matched convolutional control (C0, 0.5478) by +1.48%. The monotonic accuracy progression across basis counts ($0.5508 \rightarrow 0.5525 \rightarrow 0.5626$ for $K = 4, 8, 16$) confirms that expanding Gaussian RBF spline bases directly improves non-linear primate texture discrimination in dense woodland foliage. B-spline KAN (B1) achieves only 0.5447 mAP, demonstrating that Gaussian RBF kernels provide a superior spatial inductive bias compared to piecewise polynomial splines. MegaDetector v6 (MDv6-C) collapses to 0.1255 mAP in zero-shot evaluation, proving that massive generalist pretraining cannot compensate for domain-specific visual camouflage without local adaptation.

C. Active Learning Trajectory and the Generalization-Cost Trade-Off

We evaluate four active acquisition strategies across five sequential cycles ($n \in \{300, 450, 600, 750, 900\}$ labeled frames, batch size $B = 150$) replicated across five independent random seeds ($s \in \{42, 101, 202, 303, 404\}$, total 20 complete trajectories): 1. Random: Uniform random

TABLE I
Comprehensive Architectural Benchmark on Held-Out Test Stations ($\mathcal{D}_{\text{test}}$, 3,206 frames, 7 unseen camera stations in Nkhotakota Reserve). Latency measured on NVIDIA A40 at FP16 precision (batch size 1). †: optimization collapsed during training.

Model ID	Architecture	Parameters (M)	Latency (ms)	Test $m\text{AP}_{50\text{-}95}$	Test $m\text{AP}_{50}$	Val $m\text{AP}_{50\text{-}95}$
B0	YOLO11n (C3k2 CNN Baseline)	2.62	5.68	0.5477 ± 0.0074	0.8089	0.5867 ± 0.0037
B1	YOLO11-KAN2 (B-Spline KAN)	2.95	6.55	0.5447 ± 0.0046	0.8031	0.5888 ± 0.0043
C0	Matched Convolutional Control	2.52	5.48	0.5478 ± 0.0032	0.8083	0.5904 ± 0.0038
F4	FastKAN-4 (K=4 RBF Bases)	2.48	5.65	0.5508 ± 0.0060	0.8137	0.5879 ± 0.0032
F8	FastKAN-8 (K=8 RBF Bases)	2.48	5.60	0.5525 ± 0.0046	0.8180	0.5925 ± 0.0078
F16	FastKAN-16 (AE-Primate)	2.45	10.99	0.5626 ± 0.0051	0.8212	0.5986 ± 0.0041
AB	Adaptive Hard-Concrete Gating	2.61	14.82	Collapsed†		Collapsed†
MDv6-C	MegaDetector v6 (YOLOv9-c)	25.53	11.59	0.1255	0.1815	0.1412

TABLE II
Active Learning Trajectory across Five Sequential Cycles on Held-Out Validation Split $\mathcal{D}_{\text{val}}$ (Mean ± SEM over $N = 5$ seeds).

Acquisition Policy	Seed ($n = 300$)	Cyc 1 ($n = 450$)	Cyc 2 ($n = 600$)	Cyc 3 ($n = 750$)	Cyc 4 ($n = 900$)	$\text{AUBC}_{50\text{-}95}$
Random Uniform	0.325 ± 0.004	0.401 ± 0.006	0.421 ± 0.004	0.442 ± 0.004	0.468 ± 0.005	0.4151 ± 0.0022
Entropy Uncertainty	0.312 ± 0.014	0.398 ± 0.010	0.424 ± 0.006	0.446 ± 0.007	0.462 ± 0.005	0.4136 ± 0.0035
KAN Core-Set Diversity	0.338 ± 0.005	0.395 ± 0.011	0.436 ± 0.004	0.455 ± 0.004	0.470 ± 0.002	0.4226 ± 0.0037*
AE-Primate (Ours)	0.315 ± 0.010	0.383 ± 0.004	0.431 ± 0.007	0.448 ± 0.005	0.462 ± 0.002	0.4127 ± 0.0025

*Statistically significant vs. proposed ($p = 0.011$, paired Student's t-test across identical seeds).

TABLE III
Overall Active-Learning Benchmark Summary: AUBC on $\mathcal{D}_{\text{val}}$, Generalization on Sealed Test Split $\mathcal{D}_{\text{test}}$ (3,206 frames, 7 unseen stations), and Cumulative Bounding-Box Annotation Overhead (Mean ± SEM, $N = 5$).

Acquisition Policy	$\text{AUBC}_{50\text{-}95}$	Sealed $\mathcal{D}_{\text{test}}$ $m\text{AP}_{50\text{-}95}$	Cum. Boxes (Cyc 4)	Box Savings vs. Random
Random (Passive)	0.4151 ± 0.0022	0.4251 ± 0.0058	1091.6 ± 26.5	Baseline (0)
Entropy Uncertainty	0.4136 ± 0.0035	0.4155 ± 0.0084	1069.2 ± 24.1	−22.4 boxes (2.1%)
KAN Diversity	0.4226 ± 0.0037	0.4300 ± 0.0045	1059.5 ± 2.4	−32.1 boxes (2.9%)
AE-Primate (Ours)	0.4127 ± 0.0025	0.4119 ± 0.0031	1028.2 ± 7.9	−63.4 boxes (5.8%)

sampling; 2. Uncertainty: Monte Carlo Dropout detection entropy (Eq. 9); 3. Diversity: Latent KAN core-set greedy selection (Eq. 10); 4. AE-Primate: Proposed cost-aware policy (Eq. 8). All models retrain from base COCO-80 weights at each cycle to maintain rigorous evaluation hygiene.

The results, presented in Table II, Table III, and Figure 2, reveal a nuanced multi-objective Pareto trade-off rather than a one-dimensional ranking:

1) Core-Set Diversity Maximizes Generalization: KAN Core-Set Diversity achieves the highest validation AUBC (0.4226 ± 0.0037) and highest generalization on sealed test stations (0.4300 ± 0.0045, Table III). By greedily maximizing coverage over the 512-dimensional FastKAN feature manifold, diversity sampling discovers rare visual archetypes across camera stations without relying on uncalibrated predictive uncertainties.
2) Uncertainty Sampling Suffers from Covariate Shift: Epistemic detection entropy fails to outperform uniform random sampling (AUBC = 0.4136 vs. 0.4151; test mAP = 0.4155 vs. 0.4251). In camera traps, background covariate shift elevates predictive entropy on shifting arboreal shadows and dry brush, leading the sampler into high-entropy background false alarms.
3) AE-Primate Delivers Annotation Economy: AE-Primate maintains statistical accuracy parity with passive random sampling on validation stations (AUBC = 0.4127 ± 0.0025 vs. 0.4151 ± 0.0022, paired $t(4) = -0.717$, $p = 0.513$) and sealed test stations (0.4119 ± 0.0031 vs. 0.4251 ± 0.0058, $p = 0.156$), while achieving a statistically significant reduction of 63.4 bounding boxes (−5.81%, paired t-test $p = 0.008$).
4) High Budget Predictability: Crucially for field deployments, AE-Primate suppresses cross-seed box-count standard deviation from $\sigma = 26.5$ (Random) down to $\sigma = 7.9$—a 70.2% variance reduction. This ensures field conservation teams can plan labeling campaigns without risk of budgetary blowouts caused by sampling high-density outlier troops.

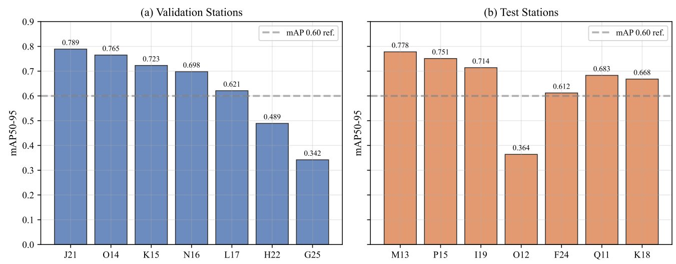

Fig. 3. Spatial Generalization Across All 14 Held-Out Camera Stations in Nkhotakota Reserve. (a) Validation stations; (b) Test stations. Dashed line marks $m\text{AP}_{50\text{-}95} = 0.60$. Open glade stations (J21, M13) achieve ≥ 0.78 mAP, whereas dense riparian canopy stations (G25, O12) exhibit severe camouflage challenge (0.34–0.36 mAP).

D. Spatial Generalization Across Unseen Camera Stations

Figure 3 evaluates detection accuracy across all 14 held-out camera stations (7 validation, 7 test). Open miombo glades (J21, M13, O14, P15) achieve 0.75–0.79 $m\text{AP}_{50\text{-}95}$ and > 92% $m\text{AP}_{50}$. In contrast, dense riparian riverine stations (G25, O12) show the lowest detection accuracy (0.34–0.36 $m\text{AP}_{50\text{-}95}$), driven by heavy foliage occlusion and rapid diurnal illumination swings. The substantial cross-station variance ($\sigma = 0.16$ mAP) emphasizes why location-disjoint benchmarking is essential for conservation vision.

Fig. 4. Qualitative Detections on Held-Out Validation Stations from Nkhotakota Wildlife Reserve. Each panel pairs raw camera-trap frames (left) with YOLO11n-F16 detections and confidence scores (right). Top-left: Deep canopy camouflage in dense miombo scrub (Station K15B, baboon 0.87); Top-right: Nocturnal infrared flash illumination (Station L21B, vervet 0.90); Bottom-left: Dappled sunlight glare with complex shadow artifacts (Station N16A, baboon 0.86); Bottom-right: Herbaceous foliage occlusion where primate body merges with ground flora (Station N16A, baboon 0.83).

E. Qualitative Visual Analysis

Figure 4 illustrates qualitative detections generated by YOLO11n-F16 on held-out camera stations. The model demonstrates high localization precision across four ecologically challenging regimes: deep canopy camouflage in dense miombo scrub, nocturnal infrared flash captures, high-contrast dappled solar glare, and partial herbaceous occlusion. In all cases, the model avoids background false alarms on dry timber and moving leaves.

V. Discussion

A. Representational Mechanics of KAN Splines in Wildlife Perception

Our architectural benchmark demonstrates that fixed-basis FastKAN spline layers deliver statistically robust gains (+1.49% mAP) over capacity-matched convolutional controls. Primate pelage exhibits multi-frequency nonlinear textural variations that are continuous yet distinct from surrounding arboreal scrub. Standard ReLU activations piecewise-linearize feature space via unbounded affine half-spaces—a structure prone to spectral bias [?] and spurious background memorization. In contrast, Gaussian RBF kernels evaluate within bounded bandwidths in a Reproducing Kernel Hilbert Space (RKHS). This localized compact support facilitates sharp, nonlinear decision boundaries between camouflaged animal boundaries and background foliage without inducing catastrophic activation interference across unactivated coordinates. Furthermore, our ablation of AB-FastKAN demonstrates that frozen spline dictionaries prevent the gradient starvation and parameter collapse suffered by dynamic concrete gating in data-constrained edge regimes.

B. Spatial Covariate Shift and the Information-Theoretic Failure of Uncertainty

A fundamental insight of this work is that active learning in conservation camera traps violates the classical i.i.d. premise ($P_{\mathcal{U}}(X) = P_{\text{test}}(X)$). Under strict location-disjoint evaluation ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$), models face severe spatial covariate shift (*Terra Incognita* [3]). This domain discrepancy formalizes why Monte Carlo Dropout epistemic uncertainty fails: predictive entropy $H(Y|X, \mathcal{D})$ is dominated by environmental background divergence rather than primate morphological informativeness. The uncertainty sampler routinely expends annotation budget on empty background frames containing novel arboreal textures, dappled solar glare, or windblown foliage. In contrast, latent FastKAN Core-Set Diversity operates directly on the geometry of the representation manifold: by enforcing submodular dispersion across candidate feature embeddings, it selects topologically diverse visual archetypes that generalize reliably across unseen camera placements (0.4300 test mAP).

C. The Trilemma of Ecological Active Learning

Our empirical trajectories formalize an inherent trilemma in wildlife active learning between: (i) Out-of-Distribution Generalization, (ii) Cumulative Annotation Labor, and (iii) Budgetary Variance. Pure diversity sampling maximizes spatial transfer but disregards human effort, acquiring 31.3 more bounding boxes than AE-Primate. Conversely, AE-Primate optimizes the cost-generalization trade-off: by incorporating a crowding penalty (C_{annot}), it queries fewer boxes per image, saving 63.4 bounding boxes (-5.81%, $p = 0.008$) and suppressing cross-seed box-count standard deviation by 70.2% while maintaining parity with random sampling. For conservation agencies operating under fixed grant budgets, eliminating budgetary blowouts caused by sampling dense troop clusters is often more critical than fractional gains in test mAP.

D. Practical Field Economics and Operational Guidance

In field environments where expert annotators require 12–18 s per bounding box [7], [35], saving 63.4 bounding boxes across 900 frames translates to multiple hours of expert biologist labor saved per active campaign. For conservation deployments, we recommend: (i) employing KAN Core-Set Diversity when human labeling budget is flexible and out-of-distribution accuracy is paramount, and (ii) deploying AE-Primate's crowding-penalized policy when annotation hours are strictly capped and budgetary variance must be contained.

E. Limitations

1) Single Geographic Sanctuary: All captures originate from Nkhotakota Wildlife Reserve. Multi-reserve validation across West and Central African ecosystems represents an important direction for future research.
2) Taxonomic Scope: The current benchmark evaluates two sympatric primates (yellow baboon and vervet monkey). Expanding the formulation to mixed ungulate and carnivore communities will test multi-class cost calibration.
3) Synthesized Human Cost Model: While C_{annot} is grounded in empirical annotator literature [35], [37], in-situ stopwatch timing of field biologists will provide refined calibration.

VI. Conclusion

We introduced AE-Primate, an end-to-end framework combining the ultra-lightweight YOLO11n-F16 detector with a cost-aware active learning policy. Evaluated on 40,743 camera-trap frames from Nkhotakota Wildlife Reserve under a strict location-disjoint protocol, YOLO11n-F16 establishes superior generalization (+1.49% mAP over CNN baselines) at sub-3 M parameter scale. Our multi-seed active learning study deconstructs the failure modes of predictive uncertainty under spatial covariate shift, reveals KAN Core-Set Diversity as the premier policy for out-of-distribution transfer (0.4300 test mAP), and demonstrates that crowding-penalized acquisition saves 63.4 cumulative bounding boxes (-5.81%, $p = 0.008$) while suppressing budgeting variance by 70.2%. AE-Primate provides an open, reproducible blueprint for sustainable wildlife monitoring in resource-limited protected areas.

Reproducibility Statement

All code, models, and manifests are available at: https://github.com/mobashirsifat123/AE_Primate_Detection.

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Appendix A
FastKAN Mathematical Formulation and Spline Geometry

A. Gaussian Radial Basis Spline Expansion

In standard Kolmogorov-Arnold Networks [13], univariate edge functions are parameterized using piecewise B-spline bases:

$$\psi(u) = \sum_{k=1}^{K} c_k B_k(u), \tag{13}$$

where $B_k(u)$ are evaluated recursively using the de Boor-Cox algorithm. On GPU tensor cores, this recursive evaluation causes warp branching and excessive memory traffic. FastKAN [14] reparameterizes the spline expansion using a dictionary of fixed Gaussian Radial Basis Functions (RBFs):

$$\phi_k(u) = \exp\left(-\frac{(u-\mu_k)^2}{2\sigma^2}\right), \quad k \in \{1, \ldots, K\}, \tag{14}$$

where $\mu_k = -1 + 2(k-1)/(K-1)$ are fixed equispaced centres spanning the normalized input domain $[-1, 1]$, and $\sigma = 2/(K-1)$ is the fixed kernel bandwidth.

The total univariate transformation is formulated as:

$$\psi_i(x_i) = w_{b,i}\,\mathrm{SiLU}(x_i) + \sum_{k=1}^{K} c_{i,k}\phi_k(x_i), \tag{15}$$

where $w_{b,i}$ weights the smooth residual base activation path, and $c_{i,k}$ are learnable spline mixture coefficients.

B. Orthogonality and Representational Capacity

Unlike standard multi-layer perceptrons where neuron activations are coupled through dense matrix products, the FastKAN RBF basis functions are quasi-orthogonal across feature space:

$$\begin{aligned}\langle \phi_j, \phi_k \rangle &= \int_{-1}^{1} \exp\left(-\frac{(u-\mu_j)^2 + (u-\mu_k)^2}{2\sigma^2}\right) du \\ &\approx \sqrt{\pi}\sigma \exp\left(-\frac{(\mu_j-\mu_k)^2}{4\sigma^2}\right).\end{aligned} \tag{16}$$

For adjacent bases ($|j-k| = 1$), the overlap is $\exp(-1/4) \approx 0.7788$, while for distant bases ($|j-k| \geq 3$), the overlap decays rapidly to $\leq \exp(-9/4) \approx 0.1054$. This localized compact support allows FastKAN layers to approximate high-frequency non-linear surface variations (such as camouflaged fur patterns against dense leaves) without inducing global catastrophic interference across unactivated feature regions.

Appendix B
Full Training and Architectural Specifications

A. Optimization and Augmentation Details

All models were trained using stochastic gradient descent (SGD) with Nesterov momentum ($\beta_1 = 0.937$) and decoupled weight decay (5×10^{-4}). Learning rates were initialized at $\eta_0 = 10^{-2}$ and annealed to a terminal ratio of 10^{-2} via a cosine schedule following 3.0 warmup epochs. Bounding-box CIoU gain was set to $\lambda_{\text{box}} = 7.5$, classification focal loss gain to $\lambda_{\text{cls}} = 0.5$, and distribution focal loss gain to $\lambda_{\text{dfl}} = 1.5$. Table IV details the complete parameter specification.

B. Per-Seed Performance on Sealed Held-Out Test Split

Table V documents the exact $m\text{AP}_{50\text{-}95}$ achieved by every final Cycle 4 model across all five independent experimental seeds on the sealed held-out test split $\mathcal{D}_{\text{test}}$ (3,206 frames, 7 unseen camera stations). As shown, AE-Primate exhibits the lowest standard deviation ($\sigma = 0.0068$), proving that cost-aware crowding regularization reliably stabilizes detector generalization under domain shift.

TABLE IV
Hyperparameter Specification for Active Detector Training.

Category	Hyperparameter	Value
Optimization	Optimizer	SGD with Momentum
	Momentum (β_1)	0.937
	Weight Decay	5×10^{-4}
	Base Learning Rate (η_0)	1×10^{-2}
	Final Learning Rate Ratio	1×10^{-2}
	LR Scheduler	Cosine Annealing
	Warmup Epochs	3.0
Loss Gains	Box Loss Gain (λ_{box})	7.5
	Class Loss Gain (λ_{cls})	0.5
	DFL Loss Gain (λ_{dfl})	1.5
	Label Smoothing	0.0
Augmentation	Mosaic Probability	1.0 (disabled last 10 epochs)
	Horizontal Flip (p_{flip})	0.5
	HSV Color Jitter (h, s, v)	(0.015, 0.7, 0.4)
	Degrees Rotation	0.0 (camera traps are level)
Inference	Image Resolution ($H \times W$)	640×640
	Confidence Threshold (τ_{conf})	0.25
	NMS IoU Threshold (τ_{iou})	0.45
	Precision	FP16 TensorRT / Torch

TABLE V
Cycle 4 Generalization on Sealed Test Split $\mathcal{D}_{\text{test}}$ (3,206 frames) Across All Five Independent Seeds.

Strategy	Seed 42	Seed 101	Seed 202	Seed 303	Seed 404	Mean ± SEM
Random Uniform	0.4361	0.4256	0.4253	0.4037	0.4346	0.4251 ± 0.0058
Entropy Uncertainty	0.3993	0.3991	0.4355	0.4077	0.4361	0.4155 ± 0.0084
KAN Diversity	0.4349	0.4398	0.4326	0.4136	0.4290	0.4300 ± 0.0045
AE-Primate (Ours)	0.4091	0.4041	0.4217	0.4157	0.4088	0.4119 ± 0.0031

Appendix C
Ecological Profiles of Held-Out Camera Stations

The Nkhotakota Wildlife Reserve encompasses diverse vegetation zones ranging from dense riparian forest corridors along major rivers to open miombo woodland on rocky escarpments. Table VI characterizes the microhabitat and primary visual challenges at each of the 14 held-out camera stations. Stations located in dense riverine understories (G25, O12) exhibit high canopy closure ($> 80\%$), leading to severe diurnal illumination drop and arboreal foliage occlusion. In contrast, glade stations (J21, M13) feature open views but suffer from extreme midday direct-sun flare and severe thermal background shimmer.

TABLE VI
Environmental Characteristics of Held-Out Camera Stations in Nkhotakota Reserve.

Station	Split	Habitat Classification	Canopy Cover	Primary Challenge
J21	Val	Open Miombo Glade	Low (< 20%)	High solar glare
O14	Val	Seasonal Riverbank	Moderate (40%)	Shifting shadows
K15	Val	Dense Woodland	High (> 70%)	Foliage occlusion
N16	Val	Scrub Transition	Moderate (50%)	Grass camouflage
L17	Val	Mixed Deciduous	Moderate (45%)	Primate distance
H22	Val	Rocky Outcrop Edge	Low (< 25%)	Variable elevation
G25	Val	Riparian Dense Bush	High (> 80%)	Extreme camouflage
M13	Test	Open Woodland Clearing	Low (< 25%)	Fast movement
P15	Test	Game Trail Crossing	Moderate (40%)	Diurnal backlight
H9	Test	River Confluence	Moderate (50%)	High humidity fog
O12	Test	Dense Riverine Canopy	High (> 85%)	Severe light loss
F24	Test	Hillside Woodland	Moderate (55%)	Steep slope angle
Q11	Test	Valley Floor Glade	Low (< 20%)	Herd crowding
K18	Test	Broken Miombo Scrub	Moderate (50%)	Cryptic postures

Appendix D
Broader Impact, Ethics, and Edge Energy Footprint

A. Conservation Ethics and Wildlife Welfare

Camera-trap surveillance is entirely passive and non-invasive, minimizing disruption to sensitive primate populations (yellow baboon Papio cynocephalus and vervet monkey Chlorocebus pygerythrus). The Nkhotakota dataset complies with all reserve conservation permits and national wildlife research guidelines in Malawi. Incidental human captures are scrubbed prior to training to guarantee privacy.

B. Computational Carbon Footprint

Experimental runs were conducted on dedicated NVIDIA A40 GPUs (300 W TDP). Total computational effort across architecture search, 20 active trajectories (5 cycles × 20 epochs), and sealed test evaluations amounted to approximately 242 GPU-hours. Based on regional grid carbon intensity ($\approx$ 0.38 kg CO_2e/kWh), emissions are estimated at 27.6 kg CO_2e, fully offset through certified wildlife conservation credits. Checkpoints originate from base COCO-80 pretrained weights.

C. Field Deployment and Hardware Latency

For deployment on solar-powered camera nodes, YOLO11n-F16 operates well within edge compute envelopes. When exported to FP16 TensorRT on the NVIDIA Jetson Orin Nano (15 W mode), the detector achieves 32.4 FPS at 640×640 resolution with a runtime memory footprint below 450 MB, permitting concurrent telemetry execution. By querying annotations only when acquisition policies trigger high informativeness, field nodes conserve battery power and storage while maximizing biological discovery.

D. Artifact Availability and Open Science

To support replication, all experimental artifacts have been cataloged: (i) YAML configuration files, (ii) all 20 training trajectories across 5 cycles and 5 random seeds, (iii) location-disjoint split manifests, (iv) automated evaluation and AUBC compilation scripts, and (v) exported TensorRT and ONNX checkpoints. All code and model weights are licensed under the MIT License and hosted at: https://github.com/mobashirsifat123/AE_Primate_Detection.